

Chapter 3

Molecular Spectroscopy and Analytical Techniques

3.1 Spectroscopy

3.1.1 Introduction

Spectroscopy deals with the interaction of atoms or molecules or matter with electromagnetic radiation. Molecules possess different quantum mechanical energy states such as electronic, vibrational, rotational and nuclear. The transition can occur between these states by the absorption or emission of radiation. These transitions are abided by quantum mechanical selection rules.

Spectroscopy involves the production, measurement and interpretation of electromagnetic spectra arising from emission or absorption of electromagnetic radiations by atoms or molecules. Detailed information regarding the structure and bonding, as well as, various inter and intramolecular processes can be obtained from the analysis of atomic and molecular spectra. The instruments used for the spectroscopic studies are known as spectrometers.

Spectroscopic methods are more reliable and accurate compared to classical methods of analysis. The following are the reasons why spectroscopic methods are widely used today

1. They require very less time for analysis
2. Less sample quantity is required
3. Sample materials remain unchanged during analysis and can be reused
4. They give accurate results

Spectroscopic techniques have been applied virtually in all fields of science and technology. Radio-frequency spectroscopy of nuclei in a magnetic field has been employed in medical imaging technique called magnetic resonance imaging (MRI) to visualize the internal soft tissue of the body with unprecedented resolution. Microwave spectroscopy is used to discover the so called three-degree blackbody radiation, the remnant of the big bang. Based on the nature of interaction, the spectra can be broadly divided into the following types

1. **Absorption spectrum:** When an atom or molecule undergo a transition from the lower energy level, E_i to the higher energy level E_f with an absorption of a photon of energy $h\nu$, the spectrum obtained is called an absorption spectrum. The important spectral techniques such as UV- visible, IR, Microwave and NMR belong to absorption spectra (fig. 3.1).

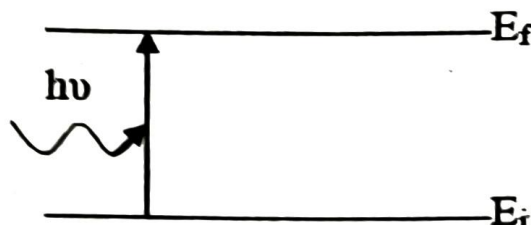


Figure 3.1: Absorption spectrum

2. **Emission spectrum:** When an atom or molecule undergo a transition from higher energy level, E_i to lower energy level, E_f with an emission of a photon of energy $h\nu$, the spectrum obtained is called an emission spectrum. For emission to be observed, higher energy states have to be more populated than lower energy states. Hydrogen spectrum, atomic emission spectrum (AES), fluorescence spectrum etc. belongs to this class (fig. 3.2).

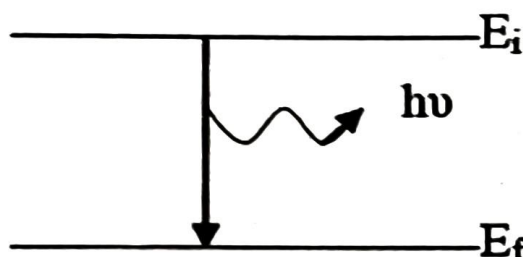


Figure 3.2: Emission spectrum

Based on the nature of the interacting species, the spectra can be divided into two categories.

1. **Atomic spectra:** Which arise from the transition of electrons between the atomic energy levels.
2. **Molecular spectra:** Which arise from the transition of electrons between different molecular energy levels. UV-visible, IR, Microwave and NMR spectra belong to this category.

3.1.1.1 Electromagnetic spectrum

Electromagnetic radiation consists of electrical and magnetic waves oscillating perpendicular to each other. They travel with velocity of light ($c = 3 \times 10^8$ m/s) in vacuum. The frequency(ν) and wave length (λ) are related by the equation, $\nu = \frac{c}{\lambda}$. Electromagnetic radiation has the

properties of both particles and waves; the particles are called photons and each possesses an amount of energy referred to as a quantum. In 1900, the German physicist Max Planck proposed that the energy of a photon (E) is directly proportional to its frequency $E = h\nu = h\frac{c}{\lambda} = hc\bar{\nu}$. The constant of proportionality h is called Planck's constant and has the value $h = 6.63 \times 10^{-34} \text{ Js}$. The SI unit of frequency is reciprocal seconds (s^{-1} or Hz). The symbol $\bar{\nu}$ stands for $\frac{1}{\lambda}$, the reciprocal of the wave length, usually expressed in spectroscopy as wavenumber in cm^{-1} . The range of photon energies is called the electromagnetic spectrum and is shown in figure 3.3. Visible light occupies a very small region of the electromagnetic spectrum, it is characterized by wavelengths from 380 nm (violet) to 750 nm (red).

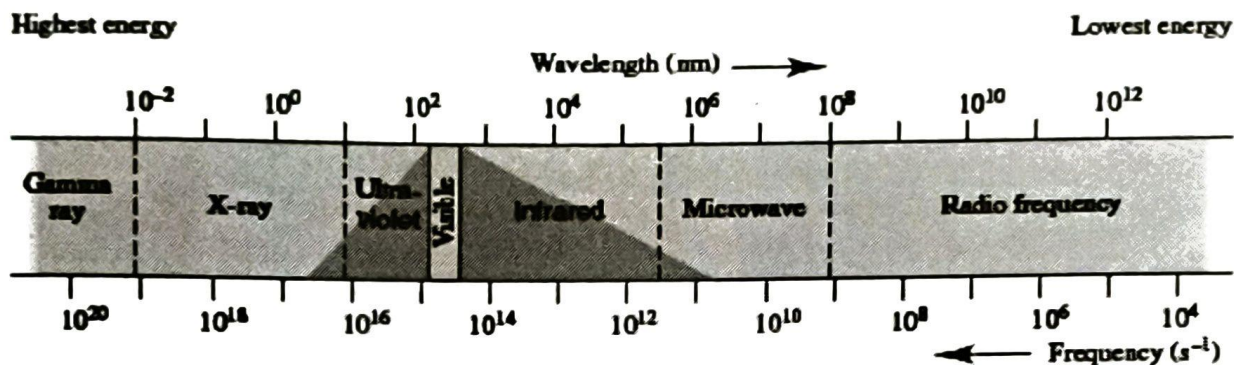


Figure 3.3: Electromagnetic spectrum

3.1.1.2 Molecular energy levels

A molecule possesses the following four types of energy.

1. **Electronic energy (E_{el}):** It is associated with the electrons present in a molecule, which can occupy various molecular energy levels. The molecular orbitals can be bonding or antibonding. The energy gaps between electronic energy levels falls in UV-visible region.
2. **Vibrational energy (E_{vib}):** Atoms in a molecule can vibrate by keeping the position of center of mass constant, the energy associated is called vibrational energy. For example HCl molecule can undergo bond stretching vibration. The energy gaps between vibrational energy levels falls in the Infrared region.
3. **Rotational energy (E_{rot}):** If a molecule rotate in space about an axis passing through its centre of mass, it is said to possess rotational energy. Rotational energy is possessed only by molecules of a gas or a liquid. The energy gaps between rotational energy levels falls in the microwave region.
4. **Translational energy (E_{trans}):** If the position of centre of mass changes with time the molecule is said to possess the translational energy. Translational energy of a molecule is usually not quantised. It becomes quantised only when the molecule is confined in a box of nano-dimension.

Therefore the total energy of the molecule can be written as the sum of all these energies

$$E_{total} = E_{el} + E_{vib} + E_{rot} + E_{trans}$$

Moreover $E_{el} \gg E_{vib} \gg E_{rot} \gg E_{trans}$ and the translational energy is negligibly small. Therefore it is approximated as $E_{total} = E_{el} + E_{vib} + E_{rot}$. Electronic transitions of a molecule are always

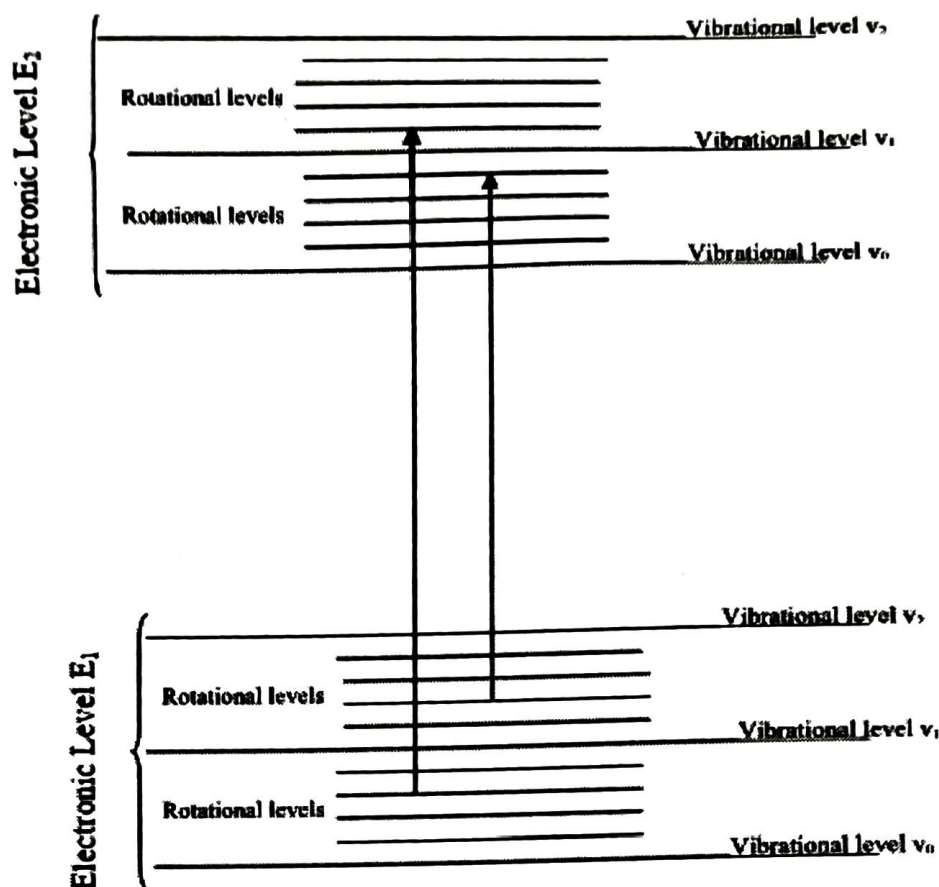


Figure 3.4: Molecular energy levels

accompanied by vibrational and rotational transitions. Similarly vibrational transitions are always accompanied by rotational transitions. Thus UV spectrum appears as broad band.

3.1.2 Beer-Lambert's law.

When a parallel beam of monochromatic electromagnetic radiation is passed through an absorbing solution of given concentration (c), the rate of decrease in intensity ($-dI$) of radiation with thickness of the solution (dx) is proportional to the intensity of incident radiation (I) at that point and also to the concentration (c) of the solution.

Note: The decrease of intensity (dI) is more when intensity of light increases. As the number of photons per unit area increases, the number of encounters of photons with absorbing molecules increases. The decrease of intensity (dI) is more when concentration of solution increases. As the absorbing molecules per unit area increases, the number of encounters of photons with absorbing molecules increases.

$$\frac{-dI}{dx} \propto Ic$$

$$\frac{-dI}{dx} = kIc \quad (3.1)$$

Where 'k' is proportionality constant.

$$\frac{-dI}{I} = kcdx$$

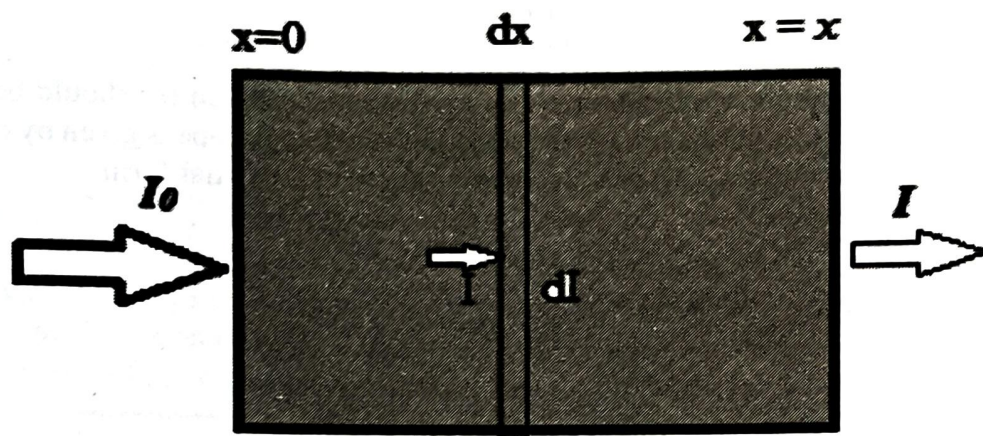


Figure 3.5: A beam of light through a solution

Let I_0 be the intensity of incident radiation and I the intensity after passing through any finite thickness x of the medium, then

$$\int_{I_0}^I \frac{-dI}{I} = \int_0^x kcdx$$

$$\ln \frac{I_0}{I} = kcx \quad (3.2)$$

$$2.303 \log \frac{I_0}{I} = kcx$$

$$\log \frac{I_0}{I} = \epsilon cx \quad (3.3)$$

Where $\epsilon = k/2.303$ called molar extinction coefficient, value of which depends on the nature of the absorbing molecule and type of absorption. The equation (3.1) differential form and equation (3.3) is the integrated form of Beer-Lambert's law. Logarithm of ratio of intensities of incident light (I_0) to the transmitted light (I) is called absorbance 'A'.

$$\log \frac{I_0}{I} = A \quad (3.4)$$

From equation (3.3) and (3.4)

$$A = \epsilon cx \quad (3.5)$$

The equation (3.5) is the simplified form of Beer-Lambert's law. Transmittance (T) is defined as the ratio of intensities of transmitted to incident light.

$$T = \frac{I}{I_0}$$

Therefore the equation (3.3) can be written as,

$$A = \epsilon cx = -\log T$$

From equation (3.5), a plot of absorbance (A) versus concentration (c) should be a straight line passing through the origin for a cell of given thickness (x), and slope is given by ϵx (fig.3.6). Beer-Lambert's law (equation (3.2)) can also be written in exponential form

$$I = I_0 e^{-kcx} \quad (3.6)$$

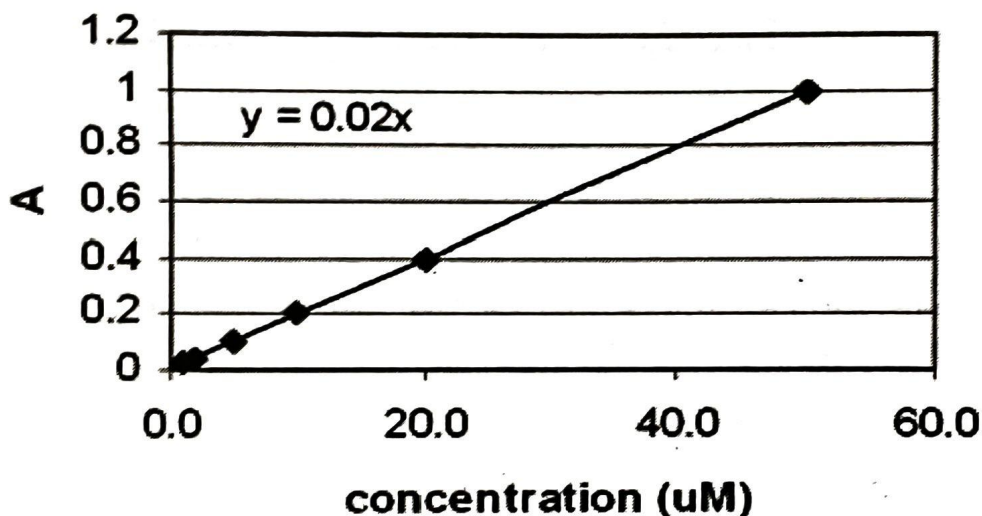


Figure 3.6: Plot of Absorbance against concentration for a given thickness

In UV-visible spectroscopy, absorbance (A) is plotted against wavelength, whereas in IR spectroscopy, transmittance is plotted against wave number.

Molar Absorptivity ϵ : From Beer-Lambert's Law $A = \epsilon cx$. Thus $\epsilon = A/cx$, where A = absorbance, c = sample concentration in moles/litre or moles/dm³ and x = length of light path through the cuvette in decimeter (dm). Thus unit of $\epsilon = \text{mol}^{-1} \text{dm}^2$

Problem 3.1.1: The percentage transmittance of a 0.01M dye solution in ethanol is 20 in a 2 cm cell for a light of wavelength 5000 Å. Find the absorbance (A) and molar absorption coefficient (ϵ).

Ans :

$$A = \log \frac{I_0}{I}$$

$$A = \log \frac{100}{20} = 0.6989$$

From equation(3.5)

$$A = \epsilon cx$$

where, $x = 2 \text{ cm} = 0.2 \text{ dm}$ Molar absorption coefficient

$$\epsilon = \frac{A}{cx}$$

$$= \frac{0.6989}{0.01 \text{ mol.dm}^{-3} \times 0.2 \text{ dm}} = 349.4 \text{ mol}^{-1} \text{ dm}^2$$

Problem 3.1.2: A dye solution of concentration 0.04 M shows absorbance of 0.045 at 530 nm; while a test solution of same dye shows absorbance of 0.022 under same conditions. Find the concentration of the test solution.

Ans:

$$A = \epsilon cx$$

Here, ϵ and thickness 'x' are constants, therefore

$$\frac{A_1}{A_2} = \frac{c_1}{c_2}$$

$$\frac{0.045}{0.022} = \frac{0.04}{c_2}$$

$$c_2 = 0.0195 \text{ M}$$

Problem 3.1.3: A 50 ppm standard solution of Fe^{3+} after developing red colour with ammonium thiocyanate shows a transmittance of 0.2 at 620 nm While an unknown solution of Fe^{3+} after developing colour with the same amount of ammonium thiocyanate gives a transmittance of 0.4. Find the concentration of unknown Fe^{3+} solution.

Ans:

$$-\log T = \epsilon cx$$

Here, ϵ and thickness 'x' are constants, therefore

$$\frac{\log T_1}{\log T_2} = \frac{c_1}{c_2}$$

$$\frac{\log 0.2}{\log 0.4} = \frac{50}{c_2}$$

$$c_2 = 28.4 \text{ ppm}$$

3.1.3 UV-visible Spectroscopy

Visible light absorption is known to all of us, because this is what causes objects to be coloured. For example, a blue dye appears blue because the light at the red end of the spectrum is absorbed, leaving the blue light to be transmitted. Human eye is functioning as a spectrometer analysing the light reflected from the surface of a solid or passing through a liquid.

Principle: Ultraviolet-Visible spectroscopy (UV = 200-400 nm, Visible = 400-800 nm) corresponds to electronic excitations between molecular energy levels. In particular, transitions involving π orbitals and lone pairs (n = non-bonding) are important and so UV-visible spectroscopy is of most use for identifying conjugated systems which tend to have stronger absorptions.

3.1.3.1 Classification of Electronic Transitions

When a molecule absorbs energy in UV-visible region, its electrons are promoted from a bonding molecular orbital to an antibonding molecular orbital. Three distinct types of electrons are involved in electronic transitions. These are σ electrons, π electrons and non-bonding electrons (n). Electronic transitions are classified into four types $\sigma \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$ and $n \rightarrow \sigma^*$ transitions. For example formaldehyde molecule has C-O σ and π bonds also non-bonding electrons over oxygen atoms.

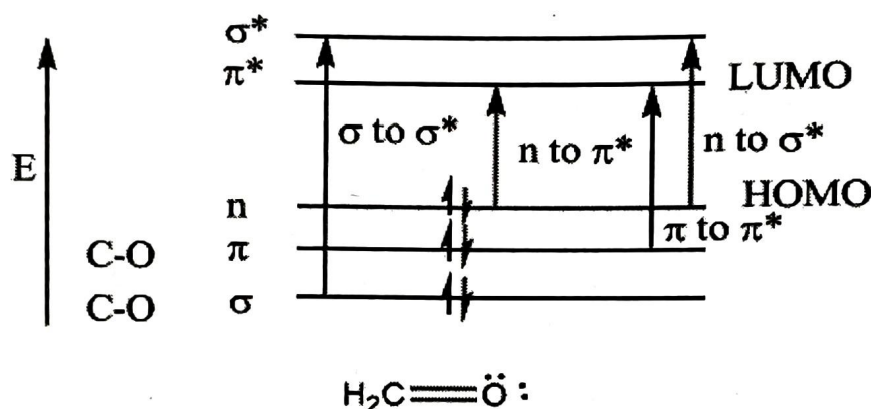


Figure 3.7: Electronic transitions in formaldehyde molecule

1. **$\sigma \rightarrow \sigma^*$ transitions:** The saturated hydrocarbons are transparent to UV-visible light. Therefore, they are colourless. The saturated hydrocarbons can give only $\sigma \rightarrow \sigma^*$ transitions of C-C σ bond and C-H σ bond. This type of transition occurs below 150 nm. The ordinary UV machines take spectra only from 200-700 nm and hence saturated hydrocarbons cannot be detected using UV-visible spectrometer.
2. **$\pi \rightarrow \pi^*$ transitions:** The unsaturated hydrocarbons containing double or triple bonds shows this type of transition. For example C = C of ethylene molecule gives an absorption maxima at 169 nm. So normal UV machines will not show such transitions. In the case of molecule containing conjugated double bonds, these transitions can be observed from highest occupied π molecular orbital (HOMO) to lowest unoccupied π molecular orbital (LUMO). In the case of 1,3-butadiene these transitions are observed at 217 nm. 1,3-butadiene has four π molecular orbitals ($\psi_1, \psi_2, \psi_3, \psi_4$) formed by the linear combination

of four atomic orbitals (p orbitals). Here ψ_1 and ψ_2 are bonding molecular orbitals and ψ_3 and ψ_4 are antibonding π molecular orbitals. In the ground state the four π electrons occupy bonding molecular orbitals. Here the transition takes place from HOMO ψ_2 to LUMO ψ_3 with $\lambda_{max} = 217$ nm.

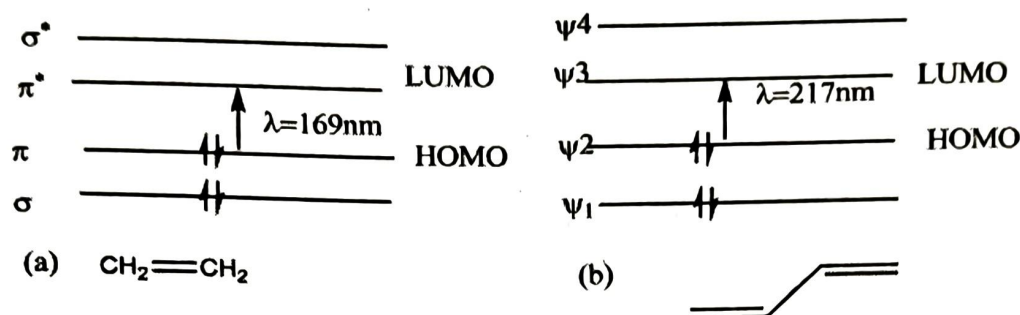


Figure 3.8: Molecular energy diagram of (a) ethene (b) 1,3-butadiene

1,3,5-hexatriene: There are three double bonds in conjugation. It has six π molecular orbitals ($\psi_1, \psi_2, \psi_3, \psi_4, \psi_5, \psi_6$), formed by the linear combination of six atomic orbitals (p orbitals). Here ψ_1, ψ_2 and ψ_3 are bonding molecular orbitals and ψ_4, ψ_5 and ψ_6 are antibonding π molecular orbitals. In the ground state the six π electrons occupy bonding molecular orbitals. Here the transition takes place from HOMO ψ_3 to LUMO ψ_4 . λ_{max} is observed at 247 nm.

Benzene: There are three double bonds in conjugation. It has six π molecular orbitals ($\psi_1, \psi_2, \psi_3, \psi_4, \psi_5, \psi_6$) formed by the linear combination of six atomic orbitals (p orbitals). Here ψ_1, ψ_2 and ψ_3 are bonding molecular orbitals and ψ_4, ψ_5 and ψ_6 are antibonding π molecular orbitals. In the ground state the six π electrons occupy bonding molecular orbitals. Here the transition takes place from HOMO to LUMO. The λ_{max} is observed at 255 nm. Here ψ_2 and ψ_3 are degenerate orbital (same energy) and ψ_4 and ψ_5 are also degenerate.

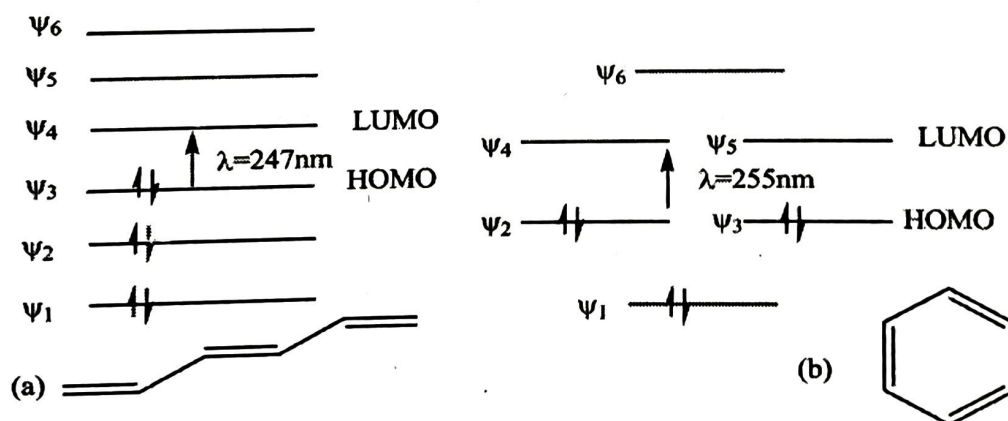


Figure 3.9: Molecular energy level diagram of (a) 1,3,5-hexatriene (b) benzene

As more and more double bonds come into conjugation the absorption comes to the visible region. In the case of lycopene a red coloured natural dye of tomato, there are 11 double bonds in conjugation therefore λ_{max} is observed at 505 nm. It is estimated that there is an increment of 30 nm with every increment of a double bond in conjugation.



Figure 3.10: Lycopene

3. **$n \rightarrow \pi^*$ transitions:** Such transitions are observed in aldehydes and ketones, which contains $>C=O$ group which has both π electrons and non bonding electrons (with oxygen atom). These transitions are generally observed with low intensity when compared with $\sigma \rightarrow \sigma^*$ and $\pi \rightarrow \pi^*$ transitions. These transitions are symmetry forbidden and hence observed with low intensity absorption. The n and π orbitals are at different planes whereas π and π^* are in the same plane. For example formaldehyde molecule shows $n \rightarrow \pi^*$ transition at 290 nm with very low intensity whereas $\pi \rightarrow \pi^*$ transition at 180 nm with high intensity (not readable in ordinary UV instrument which takes spectrum from 200 nm to 800 nm). Conjugated unsaturated aliphatic ketones show $n \rightarrow \pi^*$ transitions above 290 nm.
4. **$n \rightarrow \sigma^*$ transitions:** Compounds containing S, N, O etc which contain non-bonding electrons show these types of transitions. These transitions are also forbidden and occur in low intensity. The absorption takes place below 200 nm.

3.1.3.2 Instrumentation

The UV-visible spectrometer consists of a light source, monochromator (prism or grating), sample, reference, detector and recorder. The most suitable light sources are Tungsten Filament lamp and H-discharge lamp. In modern instruments xenon flash lamps are used. Most of the spectrometers are double beam instruments. A beam of light is split into two equal beams and one is passed through sample (transmitted intensity I) and the other passed through reference (transmitted intensity I_0). A very dilute solution of the sample is prepared using a suitable solvent and is taken in a transparent cuvette. The pure solvent is taken as reference in another similar cuvette. The most commonly used solvents are water, ethanol, hexane and cyclohexane. The intensities of these light beams are then measured by electronic detectors and compared. If the sample does not absorb light of a given wavelength then, $I = I_0$. However, if the sample absorbs light then I is less than I_0 , and this difference is detected and recorded. The graph obtained is a plot of absorbance vs wavelength.

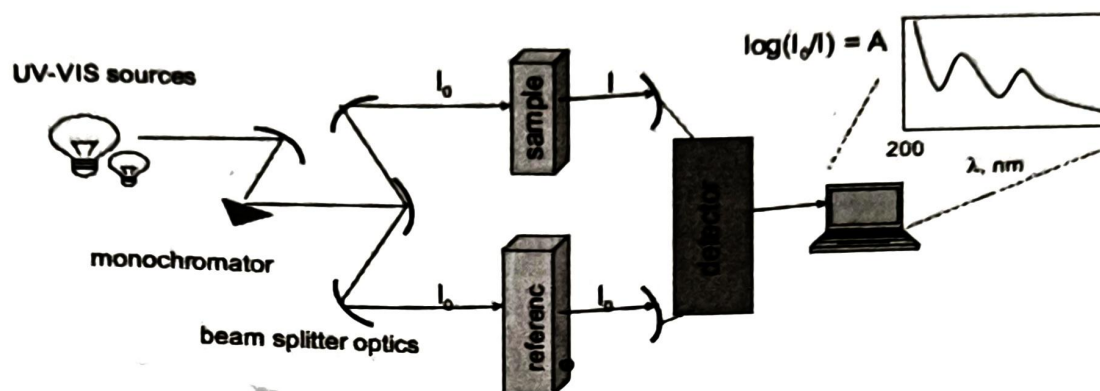


Figure 3.11: Instrumentation of UV-visible spectrometer

3.1.3.3 UV-visible spectrum

It is a plot of absorbance (A) against wavelength λ of light. The wavelength of maximum absorbance is a characteristic value, designated as λ_{max} . It is the wavelength at which absorbance is maximum. Different compounds may have very different absorption maxima and absorbance. Electronic transitions are usually accompanied by vibrational and rotational transitions. Therefore discrete lines are not observed in the spectrum and thus it appears as band with an absorption maximum.

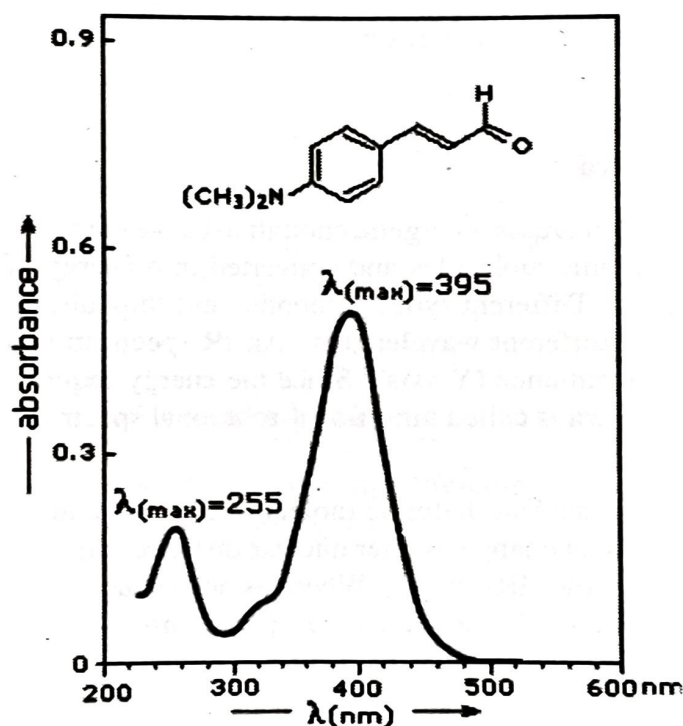


Figure 3.12: UV-visible spectrum of an organic compound

3.1.3.4 Applications of UV-visible spectroscopy

1. For the detection of aromatic compounds, conjugated dienes etc.
2. For the characterization of dyes and colorants.
3. Detection of impurities: For eg., benzene is a common impurity in cyclohexane which can be detected by the absorption at 255 nm.
4. Determination of unknown concentration: This is done by comparing the absorbance with the absorbance of the standard solution.
5. Study of kinetics of chemical reactions: If a reaction involve shift of absorption frequency or wavelength, the rate of such reaction can be easily monitored.
6. Widely used in medical lab for quantitative estimation of blood sugar, cholesterol etc. These tests are done by measuring the absorbance after developing colour with suitable reagents.

3.1.4 IR spectroscopy

Principle: Vibrational or IR spectroscopy involves the transitions between vibrational energy levels of a molecule, having change in dipole moment with vibration of a bond. These transitions are brought about by absorbing IR radiations falling in the range $500\text{--}4000\text{ cm}^{-1}$. This technique can be used to establish the identity of a compound or to determine the structure of a new compound. This technique can predict the presence of functional groups which absorb at definite frequencies.

3.1.4.1 Origin of IR spectra

The photons in the IR region are not energetic enough to cause electronic transitions. Infrared radiation is absorbed by organic molecules and converted into energy of molecular vibration, either stretching or bending. Different types of bonds, and thus different functional groups, absorb infrared radiation of different wavelengths. An IR spectrum is a plot of wavenumber (X-axis) vs percentage transmittance (Y-axis). Since the energy required is less for rotational transitions, the resulting spectra is called vibrational-rotational spectrum.

For example when a hetero nuclear diatomic molecule like HCl undergo stretching vibration the dipole moment changes with change in inter nuclear distance (dipole moment is the product of charge and distance) and shows IR activity. Where as homo nuclear diatomic molecules like H_2 , O_2 , N_2 etc. are not vibrationally active as they posses no change in dipolemoment with stretching vibration.

3.1.4.2 Number of vibrational modes in a molecule

For describing the position of an atom in a molecule it requires three degrees of freedom corresponding to three cartesian co-ordinates, therefore a molecule having n atoms, the number of degrees of freedom is equal to $3n$.

Linear molecule: For a linear molecule there are three translational and two rotational degrees of freedom, therefore total number of vibrational modes possible is equal to $3n - 5$. For a diatomic molecule ($n = 2$), thus there is only one ($3 \times 2 - 5 = 1$) vibrational mode, ie; stretching vibration. Eg. HCl can undergo one stretching vibration. Linear triatomic molecule like CO_2 has four ($3 \times 3 - 5 = 4$) vibrational modes, symmetric stretching, asymmetric stretching and two bending vibrations in two mutually perpendicular planes. Out of the four vibrational modes, only three are IR active. The symmetric stretching does not involve the change of dipole moment and is not IR active. The two bending modes are equivalent (degenerate). Therefore only two absorption frequencies are observed in the case of CO_2 , at 2349 cm^{-1} for asymmetric stretching and 667 cm^{-1} for bending vibration. For linear molecules like HCN, all the four modes are IR active.

Non Linear molecule: For a non linear molecule there are $3n - 6$ vibrational degrees of freedom. For example H_2O molecule has three atoms, hence ($3 \times 3 - 6 = 3$) vibrational freedoms. They are, symmetric stretching, asymmetric stretching and a bending mode. Unlike CO_2 , here two bending modes are not possible for H_2O . According to classical mechanics during vibration, the centre of mass should not change its position.

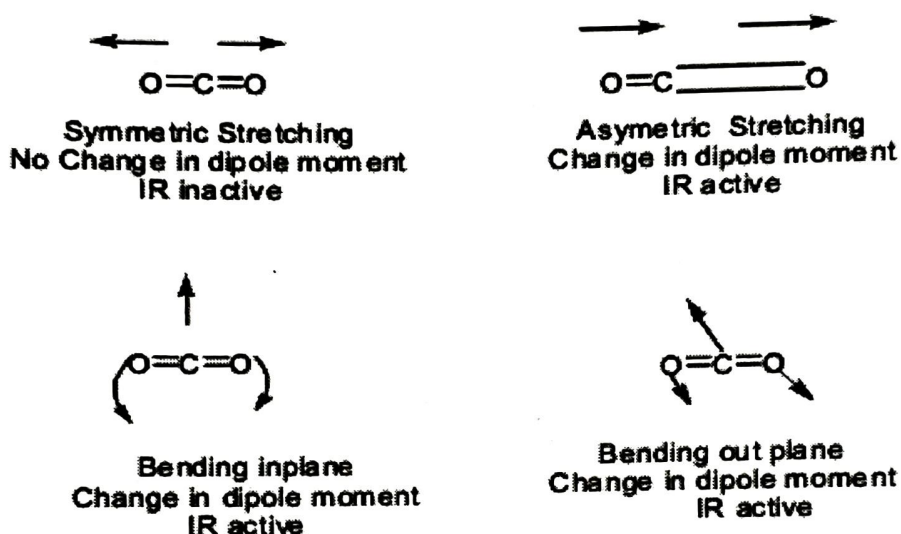


Figure 3.13: Stretching and bending vibrations of CO_2

It is to be noted that bond length changes during a stretching vibration and bond angle remains constant. But in bending vibrations bond angle changes and bond length remains constant.

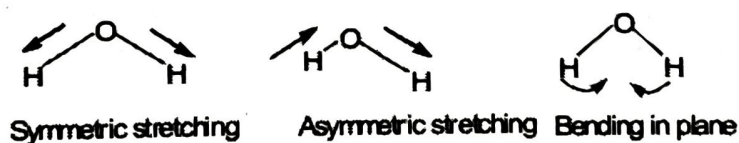


Figure 3.14: Stretching and bending vibrations of H_2O

When a group like $-NH_2$ is substituted in a heavy substrate like benzene as in aniline, more number of bending motions are observed for the $-NH_2$ group, they are rocking, scissoring, wagging and twisting.

3.1.4.3 Mechanism of interaction

During the vibration of a molecule if there occurs a change of dipole moment, it will lead to the generation of an oscillating electric field and act as a source of electromagnetic radiation. Similarly when a photon of frequency come in resonance with the frequency of vibration of a molecule, the absorption of photon takes place and the molecule starts oscillating with the frequency of radiation. Consider the asymmetric vibration of CO_2 , a fluctuating electric field produced is as shown in fig. 3.15.

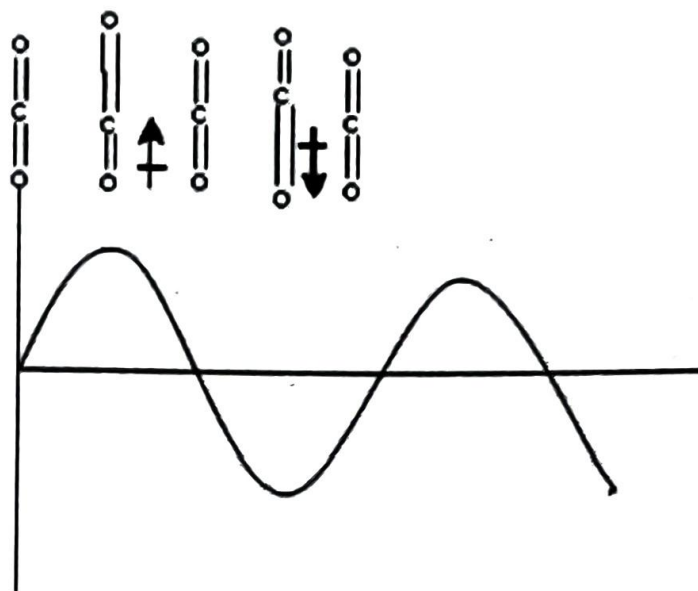


Figure 3.15: Mechanism of interaction of oscillating dipole of CO_2 in asymmetric stretching

3.1.4.4 Recording IR spectra

Earlier IR spectrometers operate in a continuous-sweep mode, whereas modern machines employ pulse Fourier-transform (FT-IR) technology. Infrared spectra can be recorded on sample regardless of its physical state; solid, liquid and gas. For a solid, the sample is mixed with potassium bromide and the mixture is pressed into a thin wafer, which is placed in the path of the infrared beam. For liquids, a drop of liquid is placed between two sodium chloride disks, through which the infrared beam is passed. The transmittance percentage is plotted against wave number in cm^{-1} .

3.1.4.5 Applications of Infrared Spectroscopy

1. **Determination of force constant of a diatomic molecule:** Fundamental frequency of vibration or wavenumber is directly proportional to square root of force constant ' k ' (bond strength) and inversely proportional to square root of reduced mass ' μ ' of the vibrating

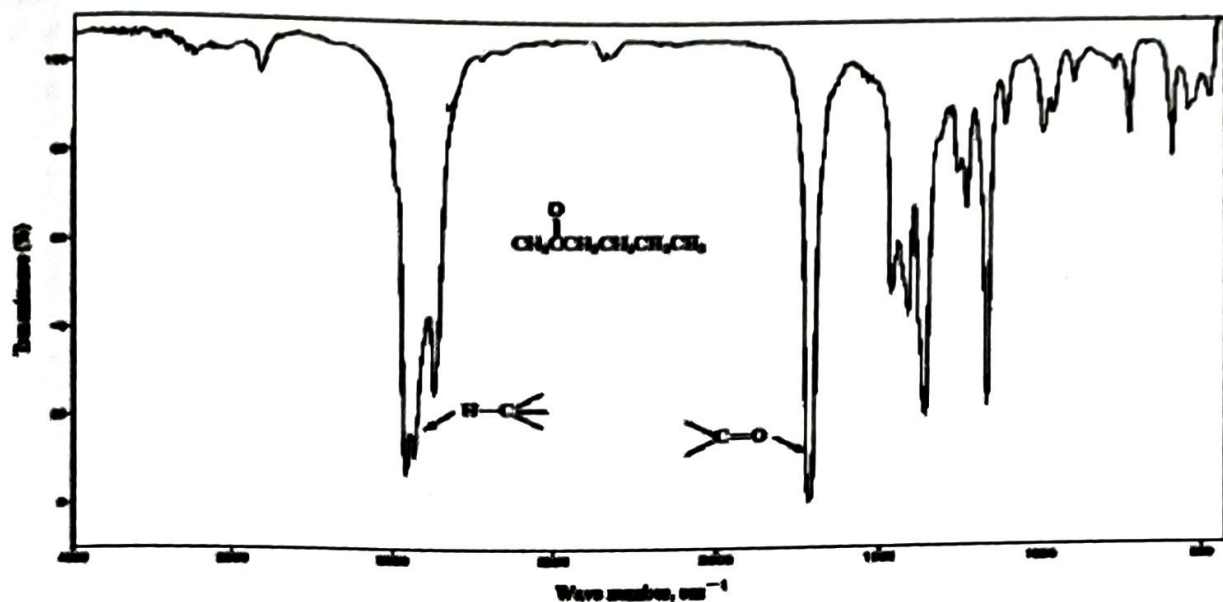


Figure 3.16: IR spectrum of 2-hexanone

atoms.

$$\text{Frequency, } \nu_0 = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

$$\text{Wave number, } \bar{\nu} = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}}$$

$$\text{Force constant, } k = 4\pi^2 \nu_0^2 \mu = 4\pi^2 c^2 \bar{\nu}^2 \mu$$

Where c is the velocity of light in vacuum (3×10^8 m/s) and μ is the reduced mass of atoms of masses m_1 and m_2 connected by a bond, $\mu = m_1 m_2 / (m_1 + m_2)$.

For example, CO molecule absorbs infrared frequency of 2140 cm^{-1} . Force constant of C≡O bond can be calculated as follows. (Given that atomic mass of C= 12 amu and O= 16 amu).

$$k = 4\pi^2 c^2 \bar{\nu}^2 \mu$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\mu = \frac{12 \times 16}{12 + 16} \times 1.66 \times 10^{-27} \text{ kg} = 1.138 \times 10^{-26} \text{ kg}$$

$$k = 4 \times 3.14^2 \times (3 \times 10^8 \text{ ms}^{-1})^2 \times (214000 \text{ m}^{-1})^2 \times 1.138 \times 10^{-26} \text{ kg}$$

$$k = 1849 \text{ Nm}^{-1}$$

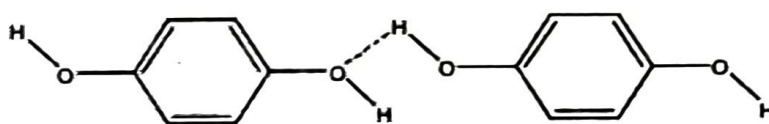
- 2. Identification of functional groups in organic molecules:** The presence of functional groups can be readily detected from the stretching vibrations of the bonds present in the functional group. For example, absorption range of >C=O group in saturated ketones is

1700-1725 cm^{-1} , -OH group in alcohol is at 3300-3500 cm^{-1} , $-\text{NH}_2$ group is at 3200-3300 cm^{-1} , C=C in alkenes is 1620-1650 cm^{-1} and C-H in organic compounds is at 2900-3000 cm^{-1} .

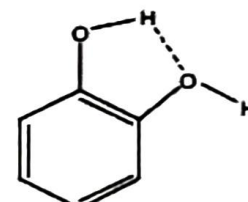
Table 3.1: Infrared absorption frequencies of some functional groups

Functional group	IR frequency (cm^{-1})	Strength of band	Bond energy (kJmol^{-1})
O-H	3200 - 3500	Strong, broad	465
N-H	3100 - 3500	Medium	390
C-H	2700 - 3100	weak	410
$\text{C}\equiv\text{C}$	2000 - 2200	weak	840
$\text{C}=\text{C}$	1600 - 1680	weak	610
$\text{C}=\text{O}$	1630 - 1800	strong	745

- Identification of unknown compounds:** This is generally done by comparison of IR spectrum of an unknown compound with that of a known compound. The comparison is especially made at the fingerprint region 700-1500 cm^{-1} which involves complex bending and fluxional modes of vibration.
- Determination of purity:** Presence of impurities can be determined by comparing the IR spectra of the given sample with the IR spectra of the authentic/pure compound.
- To distinguish between intra molecular and intermolecular hydrogen bonding:** The -OH stretching frequency (3300-3500 cm^{-1}) does not change on dilution in the case of intramolecular H-bonding. Eg; ortho-hydroxy phenol, and para-hydroxy phenol. In p-hydroxy phenol there exists inter molecular hydrogen bonding. The molecules get separated on dilution and intermolecular hydrogen bonding weakens and there is shift of absorption frequency with dilution. In o-hydroxy phenol the hydrogen bonding type is intramolecular which is not affected by dilution.



Inter molecular H-bonding in p-hydroxy phenol



intramolecular H-bonding in o-hydroxy phenol

Figure 3.17: Intermolecular and intramolecular H-bonding

3.2 Thermal Analysis

Thermal analysis deals with the study of properties of materials with change in temperature. There are different techniques available for thermal analysis which include thermogravimetric analysis (TGA), differential thermal analysis (DTA), differential scanning calorimetry (DSC), dynamic mechanical analysis (DMA), dielectric thermal analysis (DETA) etc. These methods are based on the measurement of dynamic relationship between temperature and any one of the properties such as mass change, heat change, mechanical change, phase change etc.

3.2.1 Dielectric Thermal Analysis

Dielectric analysis is a collective term used for a group of techniques that are employed for the measurement of dielectric properties of a material as a function of time, temperature or frequency. One such technique is dielectric thermal analysis (DETA), where the change in dielectric properties of a material under an applied voltage is recorded with change in temperature. It is used to study the thermal and dielectric properties of a material simultaneously and provides information about dielectric changes with temperature. It is an analogous technique to dynamic mechanical analysis (DMA) where a stress is applied instead of applied voltage. When a sinusoidal voltage is applied the sample will get polarized.

Dielectric polarisation in materials

When a material is subjected to an electric field, three types of polarisations can happen inside the material.

1. **Concentration polarisation:** When a material containing mobile ions is subjected to electric field, positive ions concentrate near the negative plate and negative ions concentrate near the positive plate. Example aluminium oxide film doped with sodium oxide, poly acrylic acid etc. Concentration polarisation is a slow process, since ions are very much heavier when compared with electrons. But polarizability is high, hence such materials possess high dielectric constant. Therefore, these types of materials can provide high capacitance when used as dielectric, for example electrolytic capacitor. These materials also possess low breakdown voltage. These types of capacitors are suitable only for low frequency, low voltage applications.
2. **Orientation polarisation:** When a material containing polar covalent bonds are subjected to an electric field, the positive ends of the dipole orient towards the negative plate, for example PVC, polyester etc. These types of polarisations are more rapid when compared to concentration polarisation as it involves no migration of ions. These materials possess moderate dielectric constant and high breakdown voltage. These types of materials are used as insulating materials for high voltage low frequency applications. PVC is generally used as insulating material in household electric wires. Polyester films are used for making capacitors for electric motors.
3. **Electronic polarisation:** In materials like polyethylene no polar covalent bonds are existing, therefore when these materials are subjected to an electric field, electron clouds around atoms lose its spherical symmetry resulting in small induced polarisation. Since electrons are very lite the polarisation takes place very rapidly. These types of materials

possess low dielectric constant and moderate breakdown voltage. Therefore, polyethylene is used as an insulating material in high frequency low voltage applications such as in radio frequency communication cables.

Principle of dielectric thermal analysis: In DETA technique the sample under study is subjected to an oscillating electric field. The applied sinusoidal voltage will polarize the sample, which forces the charge carriers inside the sample to move (positively charged particles migrate to the negative pole and vice versa). This movement results a sinusoidal current with phase shift, which is being measured. Here the polarization change of the material under varying temperatures is measured at a fixed frequency. Frequency of 1 mHz to 1 MHz is commonly used for the analysis.

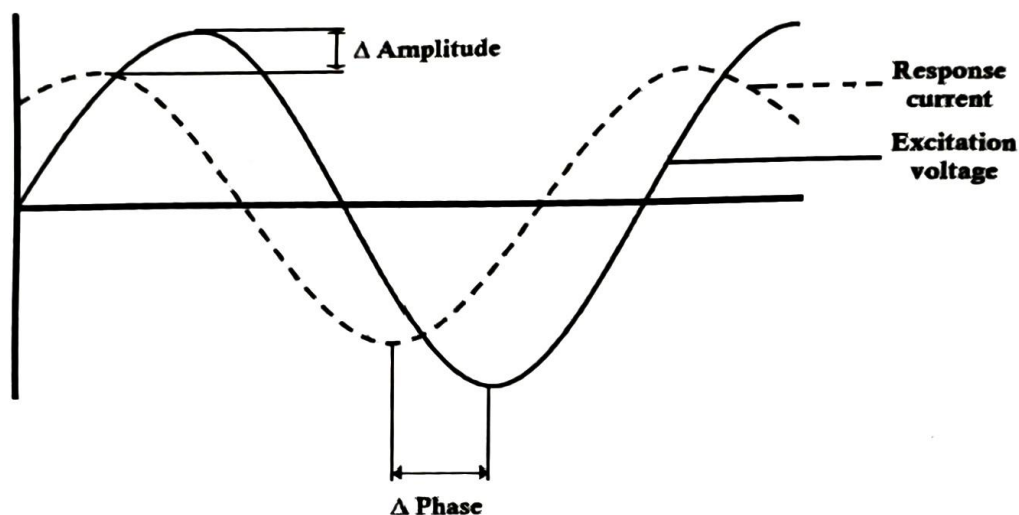


Figure 3.18: Phase shift curve

Dielectric thermal analysis measures the conductance (G , electric charge dissipated through the material) and capacitance (C , electric charge stored in the material) of a material as a function of time, temperature and frequency when it is subjected to an alternating voltage. The relaxation properties (the momentary delay of polarization response with respect to an alternating electric field) of the sample will determine the rates of conductance and capacitance. With increase in temperature relaxation time decreases due to increased molecular mobility and faster relaxation processes.

The data are generally represented in terms of the relative permittivity (ϵ') and dielectric loss factor (ϵ''). ϵ' (permittivity) is the ability of a material to store electrical energy under an applied electric field and ϵ'' (dielectric loss factor) is the energy dissipated as heat when the material is subjected to the electric field. ϵ' and ϵ'' are related to capacitance and conductance as:

$$\epsilon' = \frac{C}{\epsilon_0 \cdot (A/D)}$$

$$\epsilon'' = \frac{G}{\omega \cdot \epsilon_0 \cdot (A/D)}$$

where ϵ_0 is the permittivity of free space (8.86×10^{-12} F/m), A/D (in m), is the ratio of electrode area (A) to plate separation or sample thickness, D for a parallel plate capacitor and ω is the angular frequency.

ϵ' & ϵ'' are dimensionless quantities. The ratio ϵ''/ϵ' is the amount of energy dissipated per cycle divided by the amount of energy stored per cycle and is known as the dielectric loss tangent or dissipation factor ($\tan \delta$). It represents the efficiency of a material to convert electrical energy into heat energy under an alternating electric field. Materials with lower value of $\tan \delta$ is preferred for thermal insulator and capacitance applications.

Dielectric thermal analysis curve is generally represented as a plot of temperature in the X axis and ϵ' or ϵ'' or $\tan \delta$ in the Y axis

3.2.1.1 Instrumentation

1. **Sample holder/capacitor:** Sample is placed in between with the two parallel metal plate electrodes. Proper electrical contact should be maintained between the sample and electrodes.
A grounded electrode surrounding one of the plates, called a guard ring, is sometimes used to reduce stray electric fields to improve the performance.
2. **Programmable furnace:** Furnace with temperature programming facility is used to ramp up and down the temperature of the sample. It is used to provide the suitable heating rate also.
3. **Temperature sensor:** Thermocouples are used as the temperature sensors, which will record the temperature of the sample.
4. **Frequency generator:** Sinusoidal voltage (alternating current) is provided through the frequency generator. It allows the study of dielectric properties of the sample at different frequencies.
5. **Environment control equipment:** It is used to provide the suitable atmosphere (eg. air, inert gas) for the sample as per requirement.
6. **Temperature control equipment:** Prevents the over heating of the sample.
7. **Dielectric analyzer:** It is the key component of DETA which acts as the detector and detects the relative permittivity and dielectric loss for every change in temperature. It includes an impedance analyser or LCR meter to measure the dielectric properties of the sample.
8. **Recorder:** Computer will record the data and DETA curve is obtained as a plot between temperature and relative permittivity or dielectric loss.
Known standard materials such as low-density polyethylene (LDPE), alumina etc. are used as reference material to calibrate the system. Many DTA instruments use a dual-compartment holder for measuring the dielectric properties of both the sample and a reference material under identical conditions.

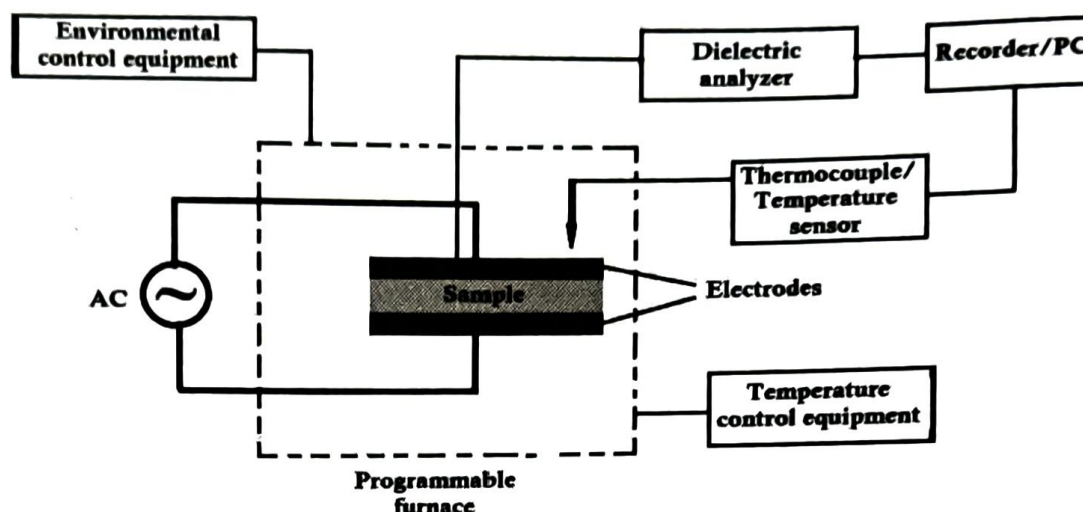


Figure 3.19: Schematic representation of DETA instrumentation

3.2.1.2 Working

Sample in the form of thin film, powder, sheet, pellet etc; is placed in between the two electrodes and an alternating electric field is applied. The temperature of the sample is increased at a control rate using programmable furnace under suitable environmental condition. The response signal is detected and analysed using dielectric analyser. In general, the sinusoidal voltage is applied to one of the electrodes and the response is then measured at the second electrode. The dielectric response is recorded for every change in temperature under a constant frequency. Temperature of the sample is monitored using the thermocouple. DETA plot is obtained between temperature vs relative permittivity or dielectric loss.

Figure 3.20 shows the DETA profile of PET at different frequencies; the glass-rubber transition of the polymer can be observed as a step increase in ϵ' and a peak in ϵ'' around 150°C .

3.2.1.3 Applications of DETA

DETA is a material characterization technique employed for polar materials in general.

1. Used to analyse food, pharmaceuticals, polymers, plastics, glasses, ceramics, proteins etc.
2. Used to identify the phase transition temperatures like glass transition temperature in polymers, especially for those which are not easily detected using other thermal technique.
3. Used to study the curing profile of thermoset, flow behaviour and phase transitions like melting, crystallization etc. in polymers, composite materials etc.
4. Used to study epoxy curing process and the related rheological properties in the fabrication of polymer-matrix composite materials.
5. Study of dielectric properties at different temperatures is used for the development of new insulating materials and energy storage materials. Materials with lower value of $\tan \delta$ is preferred for thermal insulator and capacitance applications.

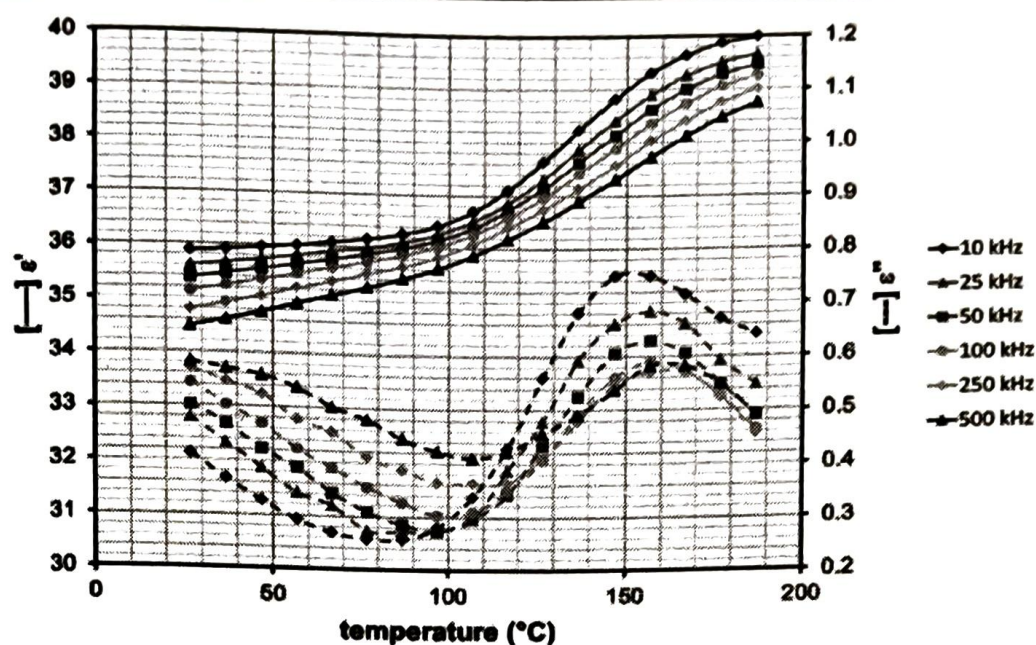


Figure 3.20: DETA curve for PET at different frequencies (Source: DOI: 10.1039/BK9781782620518-00164)

Advantages of DETA

1. It can be used to analyze solid, semi-solid and liquid materials.
2. It is more sensitive compared to other thermal analysis techniques.

3.3 Electron Microscopic Techniques

Electron microscope is a powerful magnification tool that utilizes focused beam of electrons to obtain information. Given sufficient light, the human eye can distinguish two points 0.2 mm apart, without the aid of any additional lenses. This distance is called the resolving power or resolution of the eye. Resolving power of microscopes depends on the wavelength of light used. Optical microscopes can improve the resolution only up to a magnification of 400 times, so that we can see objects of size in micro level, like bacteria. The resolution power of electron microscopy is very high (≈ 10 nm) since electrons are used for illumination (wavelength of electrons are 10^5 times shorter than that of light). Therefore nano-sized particles can be seen with the help of electron microscope. Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM) and Atomic Force Microscopy (AFM) are important surface characterization techniques used in nanotechnology.

3.3.1 Scanning Electron Microscopy (SEM)

Principle: SEM produces images of a sample by scanning the surface with a focused beam of electrons. The interaction of electron beam with the atoms near the surface of the sample results in the production different signals which include secondary electrons (SE), back scattered electrons (BSE) and X - rays. These signals are then detected by appropriate detectors and

can be used to obtain information about the surface topography and composition. The high-resolution, three-dimensional images produced by SEM makes them invaluable in a variety of science and industry applications.

Secondary electrons (SE): They are electrons that are directly knocked out from the atom by primary/incidental electron beam. They have low energy (< 50 eV) and give information about topography.

Back scattered electrons (BSE): They are incidental electrons that are reflected backwards. They have high energy (> 50 eV) compared to SE. They give information about topography and composition. BSE comes from deeper regions of the sample whereas SE originates from the surface and that's why BSE and SE carry different types of information.

X-rays: X-rays, emitted from beneath the sample surface, can provide element and mineral information.

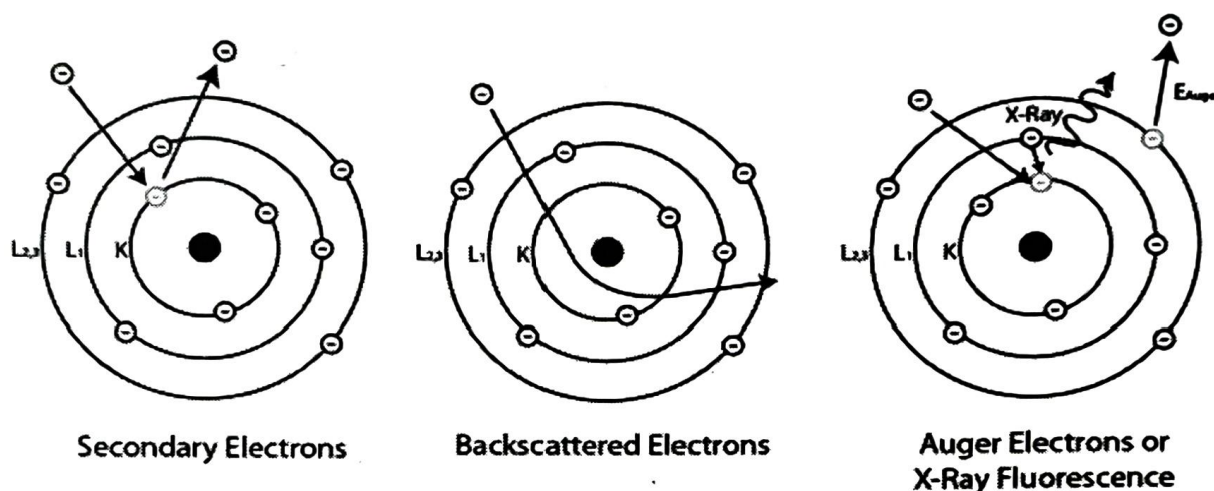


Figure 3.21: Interaction of electrons with sample

3.3.1.1 Instrumentation

SEMs consist of the following components:

1. **Electron Source:** Generates beam of energetic electrons. They are of two types:
 - (a) Thermionic gun (produces electron beam at high temperature, eg: W filament, LaB_6 etc).
 - (b) Field emission gun (produces electron beam under strong electric field).
2. **Anode:** Accelerates the electron beam. As the velocity of the electron increases wavelength decreases (according to the De Broglie relation, $\lambda = h/mV$) and thus enhances the resolution power.
3. **Vacuum chamber:** High vacuum minimises the scattering of electron beam.

4. **Condenser lens:** It is the electromagnetic lens used to focus the electron beam into small, thin and coherent beam. Generally, tubes wrapped in coils, known as solenoids, are used as condenser lenses.
5. **Scan coil:** Deflects the electron beam horizontally and vertically over the surface of the samples and scans the surface in a rectangular fashion (called raster scanning).
6. **Objective lens:** It is also an electromagnetic lens. It focuses the electron beam to the surface of the sample.
7. **Sample chamber:** The sample is kept in the sample chamber.
8. **Detectors:** There are suitable detectors to detect SE, BSE and X - rays.
9. **PC control:** To provide the SEM image.

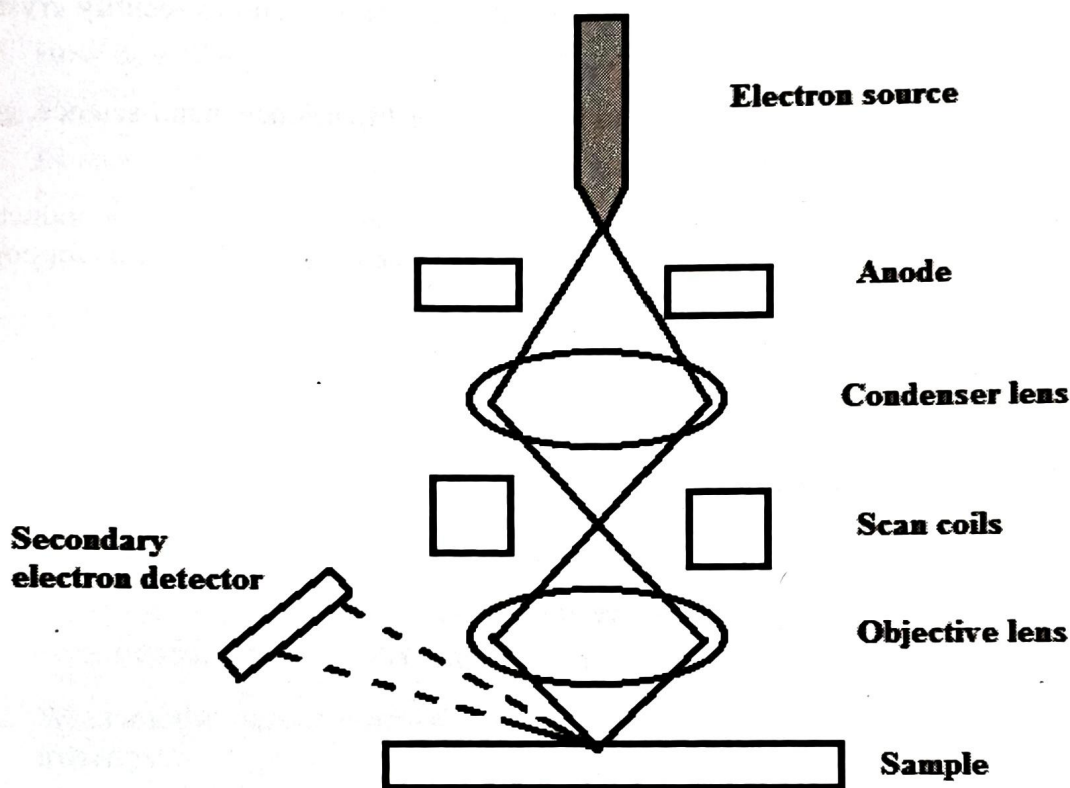


Figure 3.22: Block diagram of SEM

3.3.1.2 Working

Electron gun generates a beam of energetic electrons which are accelerated using the anode. The accelerated electrons are then focused into a coherent beam using condenser lens. Objective lens then focuses the beam onto the stage, where the solid sample is placed. Scan coil scans the surface in a raster fashion. SE, BSE and X-rays produced from the sample are detected using suitable detectors and black and white, three-dimensional images are obtained

after processing. SEM data provides information about size, shape, texture and composition of the sample. Image magnification can be up to 10 nanometers and, although it is not as powerful as its TEM counterpart.

Note: Most of the samples require some preparation before being placed in the vacuum chamber. Of the variety of different preparation processes, the two most commonly used prior to SEM analysis are sputter coating for non-conductive samples and dehydration for most biological specimens.

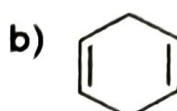
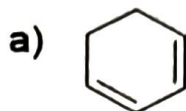
3.3.1.3 Applications

1. SEM helps in the characterizations of solid materials.
2. It can detect and analyze surface fractures, surface contamination and provide information in microstructures.
3. It helps to reveal spatial variations in chemical compositions and to identify crystalline structures.
4. It is used as an essential research tool in fields such as life science, nano-science, gemology, medical and forensic science.
5. SEM have practical industrial and technological applications such as semiconductor inspection, production line of miniscule products and assembly of microchips for computers.

3.4 Questions

3.4.1 Short answer Questions

1. Distinguish between absorption spectrum and emission spectrum.
2. What are the advantages of spectroscopic methods?
3. What are the various types of spectra?
4. Write a note on molecular energy levels.
5. State and explain Beer-Lambert's Law and differentiate between absorbance and transmittance.
6. Explain the reason for broadening of UV-Visible (electronic) spectrum.
7. How do you determine the unknown concentration of a sample using UV-visible spectroscopy?
8. Alkanes are colorless. Explain this in terms of electronic transitions possible in a molecule.
9. Give the applications of UV-visible spectroscopy.
10. Which of the following molecules can give UV-visible spectrum?
a) Methane b) Nitrogen c) Butadiene d) Benzene



11. Which molecule will absorb at longest wavelength in UV? Explain.
12. Which of the following molecules can give IR absorption? Write reason for their IR activity.
a) O_2 b) N_2 c) H_2O d) HCN e) HCl
13. Calculate the force constant of H-Cl molecule, if it gives IR absorption at 3050 cm^{-1} (atomic masses of H = 1 amu and Cl = 35 amu),
Ans: 532.8 Nm^{-1} .
14. Give any three applications of IR spectroscopy.
15. How does the IR spectroscopy help in identifying the functional groups in an organic molecule?
16. IR spectroscopy can be used to differentiate intra molecular and inter molecular hydrogen bonds. Explain with an example.
17. Explain the principle of dielectric thermal analysis.
18. Give any three applications of dielectric thermal analysis.
19. Write the principle of scanning electron microscopy.
20. List any three applications of scanning electron microscopy.

3.4.2 Essay Questions

1. Define Beer Lambert law and deduce the integrated form. Discuss the plot of absorbance verses concentration, what does the slope of the graph represents?
2. What are the various types of types of electronic transitions possible in UV-Visible spectroscopy?
3. Give the instrumentation of UV-visible spectrophotometer and explain the components in it. Comment on the role of conjugation in the wavelength of absorption with the help of examples.
4. Explain the various modes of vibration possible for CO_2 and H_2O , which of them are IR active?
5. Compare the strengths of C-H bond and C=O bond if the absorption frequencies are 3000 cm^{-1} and 1700 cm^{-1} respectively.
6. Calculate the force constant of HF molecule, if it shows IR absorption at 4138 cm^{-1} . Given that atomic masses of hydrogen and fluorine are 1 u and 19 u respectively. What would be the wavenumber if hydrogen atoms are replaced by deuterium atoms?

7. Explain the instrumentation and working of dielectric thermal analysis.
8. Explain the principle, instrumentation and working of SEM.